

Solution Requirements

Date	26 June 2026
Team ID	
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

The purpose of this section is to define the functional and non-functional requirements of the OptiCrop: Smart Agricultural Production Optimization Engine. Functional requirements describe the features and operations that the system must perform, while non-functional requirements define the quality, performance, security, and usability standards the system should meet. These requirements ensure the development of a reliable, accurate, and user-friendly AI-based crop recommendation system.

- **Functional Requirements (FR):** Define the features and operations that the OptiCrop system must perform, such as accepting soil and weather data, predicting the most suitable crop, and displaying recommendations.
- **Non-Functional Requirements (NFR):** Define the quality attributes of the OptiCrop system, including performance, reliability, security, scalability, and usability.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	Farmers can securely register and log in to access the crop recommendation system.	High

2	Authorization levels	Only authenticated users can access prediction and recommendation features.	High
3	External interfaces	The system interacts with the machine learning model, crop dataset, and web interface.	High
4	Transactions processing	The system accepts soil and weather inputs, processes them, and returns the recommended crop.	High
5	Reporting	Displays the predicted crop along with recommendation results to the farmer.	Medium
6	Business rules, etc.	Crop recommendations are generated only from valid soil and environmental input values.	High
7	Compliance to laws or regulations	User information is handled securely while following general data privacy practices.	Medium

8	Other	The system allows multiple crop predictions using different input values.	Low
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Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	The system should generate crop recommendations within a few seconds.	Prediction time < 5 seconds
2	Scalability	The application should support multiple users simultaneously.	Supports multiple concurrent users
3	Security & Data Privacy	User input data should be securely processed and protected.	Secure data handling
4	Reliability & Availability	The system should provide reliable recommendations with minimal downtime.	99% availability
5	Usability & Accessibility	The interface should be simple, responsive, and easy for farmers to use.	Easy-to-use interface

6	<i>Other</i>	The system should be easy to maintain and update as new datasets or models become available.	Maintainable architecture
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